

Competitive Programming

Section 40: Data Structures Mixed Contest

Competitive Programming Course

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1 Combining data-structure ideas

In a mixed contest, start from the operations, not the container name.

2 Complete local contest set

Problem	Title
A	Hotel Queries
B	List Removals
C	Mo Distinct Queries
D	Salary Queries
E	Functional Walk Minimum
F	Running Lower Median
G	Streaming Room Count
H	Toggle K-th Active
I	Dynamic Maximum Subarray
J	Sliding Median Cost
K	Budget Prefix Search
L	Dynamic Inversion Swaps

Problems E and F complete the direct category coverage. Problems G–L form a second, progressively harder half: each adds one modeling or invariant burden to structures already seen in Sections 36–39.

3 Operation diagnosis, not an answer key

For each local task, make an operation table:

Operation	Information needed	Online or reorderable?	Required cost
update	?	?	?
query	?	?	?

Then ask whether the answer depends on a prefix, a range aggregate, an order statistic, a first/last feasible position, or a temporary interval. The answer should determine the structure; do not select one just because it is familiar. The editorial gives the local reductions after an attempt.

4 Invariants before code

State exactly what every stored value represents. For a tree, specify the meaning of an internal node. For a frequency structure, specify which items are currently active. For offline processing, specify the current interval or set. After each update, explain why that statement remains true.

5 Complexity checks

Count operation cost over the whole input, not only one operation. A method with cheap additions can still be too slow if it rebuilds work for every query. If reordering is considered, also prove that output order can be restored and that operations commute with the reordering.

6 Exercises

Attempt A through L without looking for a named technique. For each, record the operation table, your invariant, and one naive method rejected by the constraints. Treat A–F as the core and G–L as the progressive challenge ladder. Use the editorial only after that record is complete.